Generative AI Explained

Q1

1/1 point (graded)

Generative Artificial Intelligence describes algorithms that can be used to create new, realistic content that reflects characteristics of the training data. It can produce a variety of content including audio, code, images, text, simulations, and videos. It has tremendous potential to help us become more productive. What are some applications of Generative AI?

- [x] Image Synthesis such as DALL-E or Stable Diffusion
- [x] Copilot for Software Development or Content Creation
- [x] AI-Powered Search
- [ ] Anomaly Detection
- [x] Language Modeling such as ChatGPT
- [ ] Facial Recognition

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ChatGPT and DALL-E are interfaces to the underlying Generative AI functionality. While many refer ChatGPT and DALL-E as models, this is technically incorrect. ChatGPT and DALL-E are interfaces we use to interact with the Artificial Intelligence models.

Q2

1/1 point (graded)

Unlike discriminative artificial intelligence that performs classification tasks, modern Generative Artificial Intelligence uses machine learning and deep neural networks to understand and conditionally generate new examples from complex data distributions. What are some well-established architectures used to develop Generative AI?

- [x] Embeddings to represent high-dimensional, complex data
- [x] Variational Autoecoders (VAE) use an encoder-decoder architecture to generate new data, typically for image and video generation
- [x] Generative Adversarial Networks (GAN) use a generator and discriminator to generate new data, often in video generation
- [x] Diffusion Models add and remove noises to generate quality images with high levels of detail
- [x] Transformers for Large Language Models such as GPT, LaMBDA, and LLaMa
- [x] Neural Radiance Fields (NeRF) for generating 3D content from 2D images

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Q3

1/1 point (graded)

Which of the following are ways Generative Artificial Intelligence can transform the way we work?

- [x] In education and writing, Generative AI will help us explore ideas and brainstorm
- [x] In healthcare and biology, Generative AI will help us arrive at a better understanding of our health and discover treatment options when problems arise
- [x] In agriculture, Generative AI will help us achieve greater yield by improving remote sensing and robotics
- [x] In software development, Generative AI will help us write easier and faster software by suggesting code
- [x] In geoscience, Generative AI will help us better understand Earth's ecosystems
- [ ] In manufacturing, Generative AI is used to perform defect detection
- [ ] Generative AI is used to detect cybersecurity attacks

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Q4

1/1 point (graded)

Which of the following are accurate descriptions of how Generative Artificial Intelligence works?

- [x] Generative AI models use machine learning to understand patterns in the training dataset based on probability distribution order to generate a new example based on these patterns
- [x] Generative AI models share similarities with autocomplete
- [ ] Generative AI models quickly searches through large amounts of data to simply retrieve a probable sample
- [ ] Generative AI models randomly modifies a sample from the training data to create a slightly different example

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Applications of Generative AI - Language

Q5

1/1 point (graded)

Which of the following about GPT are correct?

- [x] GPT stands for Generative Pre-Trained Transformer, which is used for training a language model to understand text or sequence of words. When it first started, GPT (2018) could perform simple text processing such as correctly identify which part of speech a particular word is in a sentence
- [x] GPT 2 (2019) could generate long form, coherent, and consistent sentences
- [x] GPT 3 (2020) demonstrated that a language model can be used for more than just understanding text and became capable of problem solving. Following its success, a series of interfaces such as WebGPT (2021), InstructGPT (2022), and ChatGPT (2022) emerged to captivate the world
- [x] At its core, Large Language Models such as those that power ChatGPT and Bard are trained at predicting next word(s), but in that process it must also learn a deep understanding of the vast language space

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Q6

1/1 point (graded)

Why is advancement in language modeling revolutionary?

- [ ] Language is compositional, which means words can be combined in order to form new meanings and expressions
- [ ] All human activities are described by language such as answering questions, solving math problems, providing instructions, using tools, and even playing games
- [ ] Language modeling has demonstrated the ability to encode, understand, and unify many different human, computer, and domain-specific languages
- [ ] Language models can be trained to use tool such as web search, run programs, navigate robots, and write code
- [x] All the above, making language models universally applicable

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Q7

1/1 point (graded)

Developing a Generative Artificial Intelligence model is resource intensive. Companies looking to leverage Generative AI have the option to either use it out-of-the-box or fine-tune them to perform specific tasks. How can enterprises efficiently fine-tune their language models to become less biased and better at solving specific tasks?

- [x] Collect and fine-tune an existing foundational model with a relatively small sample of demonstration data for specific tasks (supervised)
- [x] Use reinforcement learning and reward system to improve model outputs (RL)
- [x] Enable human feedback such as rating or ranking the model outputs (HF)
- [ ] Train a language model from scratch

✔

In machine learning, reinforcement learning from human feedback (RLHF) or reinforcement learning from human preferences is a technique that trains a "reward model" directly from human feedback and uses the model as a reward function to optimize an agent's policy using reinforcement learning (RL) through an optimization algorithm like Proximal Policy Optimization.

Q8

1/1 point (graded)

What are some examples of enterprises fine-tuning Large Language Models for specific use cases?

- [x] An LLM fine-tuned on text books can assist teachers in a classroom setting
- [x] An LLM fine-tuned on a company's call center knowledge will provide answers to questions and perform basic troubleshooting
- [x] An LLM fine-tuned on legislation documents can assist lawmakers in policy making
- [x] An LLM fine-tuned on medical literature can assist healthcare professionals in patient health diagnosis

Applications of Generative AI - Other Modalities

Q9

1/1 point (graded)

What other modalities can Generative Artificial Intelligence take?

- [x] Image-to-image to re-synthesize images and edit them
- [x] Text-to-3D to create 3D objects with a text input
- [x] Text-to-speech to synthesize audio and convert text to speech
- [x] Speech-to-speech to convert speech to speech in another voice or language

✔

Q10

1/1 point (graded)

Which industries will Generative Artificial Intelligence impact?

- ○ Service industries such as Healthcare
- ○ Geoscience
- ○ Automotive and Robotics
- ○ Virtual Worlds
- ◉ All of the above and much more

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Q11

1/1 point (graded)

How can image synthesis models be used?

- [x] Enable artistic as well as non-artistic creators to rapidly explore new ideas
- [x] Remove or lower skill or time barriers for content generation
- [x] Compose and edit images quickly, therefore increasing productivity
- [x] Alter styles, materials, textures, and even relative positions of objectives in images
- [x] Improve auto-colorization and super-resolution imaging

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Q12

1/1 point (graded)

How does Generative Artificial Intelligence and virtual worlds intersect?

- [x] Generative AI can be used to populate virtual worlds with 3D objects
- [x] Generative AI can be used to create digital avatars to represent ourselves
- [x] Generative AI can be used to create realistic environments
- [x] Generative AI can be used to create realistic user-specific interactions, engagements, and experiences
- [ ] Generative AI can be used detect cyber security attacks

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Challenges and Opportunities of Generative AI

Q13

1/1 point (graded)

What are the technical challenges of Generative Artificial Intelligence?

- [x] Developing Generative AI models with a lot of parameters involves large compute infrastructure
- [x] Building applications that use Generative AI demands fast output generation in order to be useful
- [x] Training Generative AI models is compute intensive and leads to high power consumption
- [x] Collecting training data has complicated challenges
- [x] Leveraging Generative AI technology to solve problems requires expertise

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Q14

1/1 point (graded)

Generative Artificial Intelligence produces confident, coherent results but can often be wrong. What are the common issues of Generative AI produced outputs?

- [x] Hallucination
- [x] Intellectual property ownership of AI-generated content
- [x] Cheap and easy content creation can lead to misuse, abuse, or even harm
- [x] Content produced by Generative AI can be biased based on the underlying data on which it’s trained
- [x] Realistic-sounding content can make it difficult to identify inaccuracy

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Q15

1/1 point (graded)

What are the challenges related to training data for Generative Artificial Intelligence?

- [x] High-quality training data is difficult and expensive to acquire
- [x] Data may contain biases
- [x] Some data can be harmful
- [x] Copyright license may be required
- [x] Some data should be confidential and kept private

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Q16

1/1 point (graded)

Which of the following about prompt engineering is accurate?

- [x] A prompt is the specific instruction, context, and constraints provided to the Generative AI model to achieve a task or objective
- [x] Prompt engineering involves developing a library of well-designed prompts to interface with Generative AI models
- [x] Prompt engineering involves building proper controls to create more precise, specific, and useful responses
- [x] Prompt engineering incites a new kind of creativity that comes from interacting with the Generative AI models, where one figures out how to ask questions in more useful ways to get valuable results
- [ ] Building a language model from scratch is more efficient than prompt engineering for fine-tuning Generative AI models

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Summary

Q17

1/1 point (graded)

How should Generative Artificial Intelligence models be evaluated?

- [x] Quality: Especially for applications that interact directly with users, having high-quality generation outputs is key. For example, in speech generation, poor speech quality is difficult to understand. Similarly, in image generation, the desired outputs should be visually indistinguishable from natural images
- [x] Diversity: A good generative model captures the minority modes in its data distribution without sacrificing generation quality. This helps reduce undesired biases in the learned models
- [ ] Quantity: A generative model should aim to be used as much as possible because generating content is cheap and easy
- [x] Speed: Many interactive applications require fast generation, such as real-time image editing to allow use in content creation workflows

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1/1 point (graded)

What are ways to improve Generative Artificial Intelligence?

- [x] Train models that are more restricted but still useful
- [x] Carefully consider how tools are deployed and build safeguards so that the models are more controllable
- [x] Align models with human preferences so that they are safer to use
- [x] Optimize training speed and cost through improvements in both hardware and software
- [x] Enable accessibility and flexibility for developers
- [x] Reduce time to generate outputs

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Q19

1/1 point (graded)

What technologies have propelled the advancements in Generative Artificial Intelligence?

- [x] GPU computing that enables parallel processing of large datasets (scale up)
- [x] Network capabilities that enable distributed systems to work in parallel (scale out)
- [x] Software that enables developers to leverage powerful hardware
- [x] Cloud services that enable access to scarce and expensive hardware
- [x] Innovation in neural network architectures

Q20

1/1 point (graded)

What are the benefits of Generative Artificial Intelligence?

Note: Written by the Generative AI model ChatGPT.

- [x] Generative AI algorithms can be used to create new, original content, such as images, videos, and text. This can be useful for applications such as entertainment, advertising, and creative art
- [x] Generative AI algorithms can be used to improve the efficiency and accuracy of existing AI systems, such as natural language processing and computer vision. For example, Generative AI algorithms can be used to create synthetic data that can be used to train and evaluate other AI algorithms
- [x] Generative AI algorithms can be used to explore and analyze complex data in new ways, allowing businesses and researchers to uncover hidden patterns and trends that may not be apparent from the raw data alone
- [x] Generative AI algorithms can help automate and accelerate a variety of tasks and processes, saving time and resources for businesses and organizations

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